University
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College of
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Deliverable #: lab 5 (logical design - normalization)

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
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Team Information

Team Name	Role
Yassine TAHA	Task 1 (first part) + Task 2(a)
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1 Task 1:

After examining the functional dependencies specified in the Lab 4 solution, we found no relations that violate BCNF. For every stated non-trivial functional dependency, the determinant (left-hand side) is a candidate key or a superkey of its relation. Relations without non-trivial functional dependencies also satisfy BCNF. Therefore, based on the functional dependencies explicitly specified in Lab 4, no decomposition is required for Task 1.

2 Task 2 :

2.1 a)

- $\text{serialNo}^+ = \{\text{serialNo, modelId, manufacturer, labId}\}$, so serialNo is a candidate key.
- The relation does not satisfy BCNF.
- $\text{modelId} \rightarrow \text{manufacturer}$ violates BCNF because modelId is not a superkey, since $\text{modelId}^+ = \{\text{modelId}, \text{manufacturer}\}$.
- The relation also violates 3NF: manufacturer is a non-prime attribute that depends transitively on the key through modelId. Since the only key serialNo is a single attribute, there is no partial dependency, so the strongest normal form is 2NF.
- $R_1(\text{modelId}, \text{manufacturer})$, $R_2(\text{serialNo}, \text{modelId}, \text{labId})$; both are in BCNF.
- The decomposition is lossless, because $R_1 \cap R_2 = \{\text{modelId}\}$ and $\text{modelId} \rightarrow R_1$.
- Both original FDs are preserved: $\text{modelId} \rightarrow \text{manufacturer}$ in R_1 and $\text{serialNo} \rightarrow \{\text{modelId}, \text{labId}\}$ in R_2.

2.2 b)

- $(\text{pid}, \text{projectCode})^+ = \{\text{pid, projectCode, role, piPid}\}$, so $\{\text{pid}, \text{projectCode}\}$ is a candidate key. It is minimal because $\text{pid}^+ = \{\text{pid}\}$ and $\text{projectCode}^+ = \{\text{projectCode}, \text{piPid}\}$. It is the only candidate key.
- The relation does not satisfy BCNF.
- $\text{projectCode} \rightarrow \text{piPid}$ violates BCNF because projectCode is not a superkey, since $\text{projectCode}^+ = \{\text{projectCode}, \text{piPid}\}$ does not contain pid or role. The other FD, $(\text{pid}, \text{projectCode}) \rightarrow \text{role}$, causes no violation because its determinant is the candidate key.
- piPid is a non-prime attribute and depends on only part of the composite key (a partial dependency), so the relation violates 2NF. It is only in 1NF (assuming atomic values), so the strongest normal form is 1NF.
- $R_1(\text{projectCode}, \text{piPid})$, $R_2(\text{pid}, \text{projectCode}, \text{role})$; both are in BCNF, since projectCode is the key of R_1 and $(\text{pid}, \text{projectCode})$ is the key of R_2.

- The decomposition is lossless, because $R_1 \cap R_2 = \{\text{projectCode}\}$ and projectCode $\rightarrow R_1$.
- Both original FDs are preserved: projectCode $\rightarrow$ piPid in R_1 and (pid, projectCode) $\rightarrow$ role in R_2.

2.3 c)

- $(\text{modelId}, \text{certCode})^+ = \{\text{modelId, certCode, certTitle, issuingAuthority}\}$, so {modelId, certCode} is a candidate key. It is minimal because $\text{modelId}^+ = \{\text{modelId}\}$ and $\text{certCode}^+ = \{\text{certCode}, \text{certTitle}, \text{issuingAuthority}\}$. Since neither modelId nor certCode appears on the right-hand side of any FD, both must belong to every key, so it is the only candidate key. This matches the many-to-many relationship between equipment models and certifications.
- The relation does not satisfy BCNF.
- certCode $\rightarrow \{\text{certTitle}, \text{issuingAuthority}\}$ violates BCNF because certCode is not a superkey, since $\text{certCode}^+ = \{\text{certCode}, \text{certTitle}, \text{issuingAuthority}\}$ does not contain modelId. As a result, the title and issuing authority of a certification are repeated for every model that requires it.
- certTitle and issuingAuthority are non-prime attributes and depend on only part of the composite key (a partial dependency), so the relation violates 2NF. It is only in 1NF (assuming atomic values), so the strongest normal form is 1NF.
- $R_1(\text{certCode}, \text{certTitle}, \text{issuingAuthority})$, $R_2(\text{modelId}, \text{certCode})$; both are in BCNF, since certCode is the key of R_1 and R_2 is all-key with no non-trivial FD.
- The decomposition is lossless, because $R_1 \cap R_2 = \{\text{certCode}\}$ and certCode $\rightarrow R_1$.
- The FDs are preserved: certCode $\rightarrow \{\text{certTitle}, \text{issuingAuthority}\}$ is in R_1, and (modelId, certCode) remains the key of R_2. The dependency (modelId, certCode) $\rightarrow \{\text{certTitle}, \text{issuingAuthority}\}$ follows from certCode $\rightarrow \{\text{certTitle}, \text{issuingAuthority}\}$ by augmentation, so it does not need to be checked separately.

2.4 d)

- $(\text{suppId}, \text{consId}, \text{labId})^+ = \{\text{suppId, consId, labId, supplierName, unitPrice}\}$, so {suppId, consId, l is a candidate key. Since none of these three attributes appears on the right-hand side of any FD, all three must belong to every key, so it is the only candidate key.
- The relation does not satisfy BCNF.
- suppId $\rightarrow$ supplierName violates BCNF because suppId is not a superkey, since $\text{suppId}^+ = \{\text{suppId}, \text{supplierName}\}$ does not contain consId, labId or unitPrice. The other FD, (suppId, consId, labId) $\rightarrow$ unitPrice, causes no violation because its determinant is the candidate key.
- supplierName is a non-prime attribute and depends on only part of the composite key (a partial dependency), so the relation violates 2NF. It is only in 1NF (assuming atomic values), so the strongest normal form is 1NF.

- $R_1(\text{suppId}, \text{supplierName})$, $R_2(\text{suppId}, \text{consId}, \text{labId}, \text{unitPrice})$; both are in BCNF, since suppId is the key of R_1 and $(\text{suppId}, \text{consId}, \text{labId})$ is the key of R_2.
- The decomposition is lossless, because $R_1 \cap R_2 = \{\text{suppId}\}$ and $\text{suppId} \rightarrow R_1$.
- Both original FDs are preserved: suppId $\rightarrow$ supplierName in R_1 and $(\text{suppId}, \text{consId}, \text{labId}) \rightarrow$ unitPrice in R_2.

2.5 e)

- $\text{resId}^+ = \{\text{resId, projectCode, projectTitle, madeByPid, status}\}$, so resId is a candidate key. Since resId^+ encompasses all attributes in the relation, it is the only candidate key.
- The relation does not satisfy BCNF.
- projectCode $\rightarrow$ projectTitle violates BCNF because projectCode is not a superkey, since $\text{projectCode}^+ = \{\text{projectCode, projectTitle}\}$ does not contain resId, madeByPid, or status.
- The relation also violates 3NF: projectTitle is a non-prime attribute that depends transitively on the candidate key through projectCode. Since the sole candidate key resId is a single attribute, there are no partial dependencies, meaning the strongest normal form satisfied is 2NF.
- $R_1(\text{projectCode}, \text{projectTitle})$, $R_2(\text{resId}, \text{projectCode}, \text{madeByPid}, \text{status})$; both are in BCNF, since projectCode is the key of R_1 and resId is the key of R_2.
- The decomposition is lossless, because $R_1 \cap R_2 = \{\text{projectCode}\}$ and projectCode $\rightarrow R_1$.
- Both original FDs are preserved: projectCode $\rightarrow$ projectTitle is in R_1 and resId $\rightarrow \{\text{projectCode}, \text{madeByPid}, \text{status}\}$ is in R_2.

3 Project Integration Report

2. Key Challenges Encountered and Resolutions

- **Candidate keys and BCNF violations:** we used attribute closures to find the candidate keys and checked whether the determinant of each functional dependency is a superkey.
- **Strongest normal form:** it was initially hard to separate 1NF, 3NF and BCNF violations. We resolved this by applying the definitions strictly, checking each FD against the candidate keys.
- **Lossless decomposition:** we checked that the intersection of the decomposed relations is always a superkey of at least one of them, and we verified dependency preservation after each decomposition.

3. Justification of Conversion Choices

- **Task 1:** we used only the functional dependencies specified in the Lab 4 solution, so no decomposition was needed.
- **Task 2:** each BCNF decomposition isolates the violating FD in a new relation whose key is the determinant. The determinant stays in the original relation as a foreign key, which guarantees a lossless join. The new relations are:
  - (a) Model(modelId, manufacturer)
  - (b) Project(projectCode, piPid)
  - (c) Certification(certCode, certTitle, issuingAuthority)
  - (d) Supplier(suppId, supplierName)

  These choices eliminate the BCNF violations, and all decompositions are lossless and dependency-preserving.

4. Assumptions Made

- We considered only the FDs provided in the lab statement, assumed they are complete, and added no other dependencies.
- All attributes are atomic, so every relation is in 1NF.
- For Task 2(c), the composite key (modelId, certCode) models the many-to-many relationship, with no hidden FDs beyond those stated.